

Automation, Offshoring, and Wage Inequality in Japan*

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Abstract

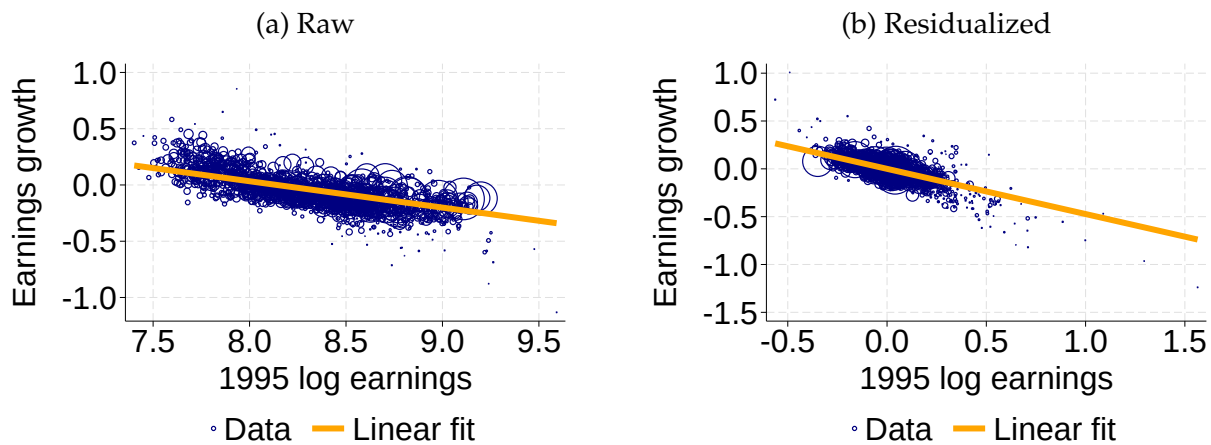
Between-group earnings inequality among full-time regular employees in Japan narrowed between 1995 and 2018. I study whether exposure to automation and offshoring contributed to this convergence using 1,504 demographic groups defined by prefecture of work, sex, age, and education. Automation has little association with earnings growth. By contrast, greater offshoring is concentrated among initially lower-earning groups and is associated with faster subsequent earnings growth. IV estimates indicate that a one-standard-deviation increase in offshoring raises monthly earnings by about 0.048 log points, almost entirely through hourly wages, rather than hours worked.

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1 Introduction

Earnings differences across groups of full-time regular employees in Japan narrowed substantially between 1995 and 2018. Among 1,504 groups defined by prefecture of work, sex, age, and education, the weighted standard deviation of log monthly earnings fell from 0.36 to 0.29. Figure 1 shows the same convergence directly: groups with higher earnings in 1995 experienced lower subsequent earnings growth. The relationship remains after removing Chinese import exposure and additive prefecture and demographic fixed effects.

Figure 1: Convergence in Monthly Earnings



Notes: This figure shows the 1995–2018 change in log mean real monthly earnings against log mean real monthly earnings in 1995. Panel (a) shows the raw relationship, and panel (b) shows the residualized relationship. Navy circles show the 1,504 groups. Circle areas are proportional to 1995 total hours, and orange lines show hours-weighted linear fits. Both residualization and the linear fits use 1995 total hours as analytic weights. Panel (b) residualizes both axes on China exposure, the group’s 1995 wage-bill share in industries for which the China shock is defined, and prefecture, sex, five-year age-group, and college fixed effects.

In this paper, I study whether exposure to automation and offshoring can explain that convergence. Automation and offshoring can both reallocate tasks away from domestic workers, but their effects on earnings can theoretically go either way. Automation displaces labor from tasks while potentially lowering production costs and raising labor demand elsewhere. Offshoring moves tasks abroad while giving domestic producers access to imported inputs, which can increase labor demand. The incidence of each force depends on the industries in which workers are employed and the tasks they perform. Japan is a useful setting because its full-time workforce experienced convergence while its production structure became more automated and more integrated with foreign input markets.

Following [Acemoglu and Restrepo \(2022\)](#), I regress changes in earnings on automation and offshoring measures, using shift-share designs. Automation exposure combines the 1995–2018 decline in sectoral labor share with the group’s 1995 wage-bill concentration in routine occupations. Offshoring exposure combines supplier import penetration, input purchases, domestic materials costs, and the group’s initial wage-bill concentration in offshorable occupations.

The results differ sharply across the two exposures. Automation has little association with earnings growth, in contrast to the results for the United States ([Acemoglu and Restrepo, 2022](#)). Offshoring is associated with convergence. The groups with initially lower earnings are more exposed to offshoring and subsequently experienced faster earnings growth. IV estimates using foreign trade growth outside Japan and China yield an earnings coefficient of 1.31. A one-standard-deviation increase in offshoring corresponds to about 0.048 log points of earnings growth.

The IV results also indicate that the earnings response operates through aggregate hourly wages rather than monthly hours. For a one-standard-deviation increase in offshoring, the estimates imply approximately 0.051 log points of wage growth, so increases in hourly wages account for almost all of the earnings response.

Related literature. First, I build on the task approach to technological change and wage inequality. [Autor et al. \(2003\)](#) document that computerization reduced demand for routine tasks. [Acemoglu and Autor \(2011\)](#) and [Acemoglu and Restrepo \(2018\)](#) organize these changes through the assignment of tasks across labor and capital. The empirical framework I use is [Acemoglu and Restrepo \(2022\)](#), who measure task displacement across U.S. demographic groups and show that automation-driven task displacement contributed to rising wage inequality. I apply this demographic-group framework to Japan, compare automation with offshoring, and distinguish changes in monthly earnings from changes in aggregate hourly wages and hours.

Second, this paper relates to evidence on wage inequality and technological change in Japan. Broader studies document changes in wage inequality and in the distributions of household earnings, income, and wealth ([Ohtake, 2005](#); [Kambayashi et al., 2008](#); [Kitao and Yamada, 2025](#); [Sano, 2026](#)). Other work emphasizes the role of college-educated labor supply and the divide between regular and nonregular work ([Kawaguchi and Mori, 2016](#); [Kambayashi, 2017](#)). [Ikenaga and Kambayashi \(2016\)](#) document employment polarization and find an industry-level association between ICT capital penetration and declines in routine-task intensity. [Dekle \(2020\)](#) finds that robot-adopting industries did not reduce labor demand, and [Adachi et al. \(2024\)](#) find no negative impact on local employ-

ment rates. [Kikuchi et al. \(2024\)](#) show that automation shifts employment from routine manufacturing occupations toward services, especially among younger and non-college workers. [Jiang and Sano \(2024\)](#) survey evidence on how population aging and automation affect employment and wage differences across groups. I study how automation and offshoring relate to earnings growth across groups of full-time regular workers.

Finally, I contribute to the literature on trade, offshoring, and labor-market outcomes. [Feenstra and Hanson \(2003\)](#) review evidence linking global production sharing to rising wage inequality and summarize measures based on purchased imported inputs. [Autor et al. \(2013\)](#) document adverse local-labor-market effects of Chinese import competition. By contrast, [Ottaviano et al. \(2013\)](#) and [Wright \(2014\)](#) find that productivity gains from offshoring can dominate its displacement effect and increase the relative demand for exposed workers.¹ I separate access to imported inputs (offshoring) from Chinese import competition and estimate how offshoring relates to earnings, aggregate hourly wages, and hours across Japanese demographic groups.

2 Empirical Specification and Data

2.1 Empirical Specification

Following the direct task-displacement measure in [Acemoglu and Restrepo \(2022\)](#), I write the automation benchmark as

$$X_g^A = \sum_{i \in \mathcal{I}} \omega_g^i \frac{\omega_{gi}^R}{\omega_i^R} \left(-d \log s_i^L \right), \quad (1)$$

where ω_g^i is group g 's initial wage-bill share in industry i , $\omega_{gi}^R / \omega_i^R$ measures the group's relative concentration in routine work within that industry, and $-d \log s_i^L$ is the labor-share decline.

The offshoring counterpart is

$$X_g^O = \sum_{i \in \mathcal{I}} \omega_g^i \frac{\omega_{gi}^O}{\omega_i^O} \left(-d \log (1 - m_i) \right), \quad (2)$$

where $d \log x_i \equiv \log x_{i,2018} - \log x_{i,1995}$, m_{it} is the imported share of materials costs in industry i and year t , and $\omega_{gi}^O / \omega_i^O$ measures the group's relative concentration in off-

¹[Endoh \(2021\)](#) studies working-hours adjustment within Japanese firms, and [Sasahara \(2022\)](#) reviews the empirical China-shock literature.

shorable work within that industry. Thus, equation (2) weights industries by offshorable-task intensity in the same way that equation (1) weights industries by routine-task intensity, while the industry shock is the log decline in the domestic share of materials costs.

For exposure $k \in \{A, O\}$, the preferred specification is

$$\Delta y_g = \alpha + \beta_k X_g^k + \rho_k S_g^k + \gamma C_g + \rho_C S_g^C + \eta_{\text{pref}(g)} + \eta_{\text{demo}(g)} + \varepsilon_g, \quad (3)$$

where X_g^k is automation or offshoring. Let q_{gi}^k denote the corresponding exposure weight and v_i^k indicate that industry i has a valid shock. The exposure-specific incomplete-share control is $S_g^k = \sum_i q_{gi}^k v_i^k$. Likewise, C_g is China exposure and $S_g^C = \sum_i q_{gi}^C v_i^C$ is its incomplete-share control, where q_{gi}^C is the group's initial industry wage-bill share and v_i^C indicates a valid China shock.

2.2 Data

Basic Survey on Wage Structure (BSWS) The BSWS provides worker earnings, hours, employment, industry, and occupation (Ministry of Health, Labour and Welfare, 2018). The main sample uses 1995 and 2018, and the pre-period check uses 1980 and 1995. I keep private-sector, full-time regular employees aged 25–64. I form 1,504 groups from 47 prefectures, two sexes, eight five-year age groups, and college status. Survey expansion weights determine group totals and means. Monthly earnings equal contractual cash earnings plus one-twelfth of annual special cash earnings, and monthly hours equal scheduled plus overtime hours. I express earnings in 2015 yen using the Japan CPI (Organization for Economic Co-operation and Development, 2023). The aggregate hourly wage equals survey-weighted total monthly earnings divided by survey-weighted total monthly hours within a group. The 1995 industry and occupation wage bills determine the initial group shares used to construct the exposures.

Japan Industrial Productivity (JIP) Database The JIP 2023 Database provides data on annual sector accounts (Research Institute of Economy, Trade and Industry and Hitotsubashi University, 2023). I use labor costs, nominal value added, and net taxes in 1995 and 2018 to construct industry labor shares. I also use the 1995 and 2018 input–output tables, which report purchases from each supplier product by each buyer industry. These purchases allocate imported inputs across buyer industries, and JIP nominal output provides the corresponding aggregation weights.

Census of Manufacture (CoM) The 1995 and 2018 public-use CoM industry tables provide domestic shipments and materials costs for manufacturing industries (Ministry of Economy, Trade and Industry, 2019). The preferred offshoring measure uses three-digit statistics for establishments with at least four persons engaged in both years. Shipments provide the domestic-production component of Japanese absorption and the weights used to map supplier products to manufacturing industries. I use materials costs and combine them with imported-input purchases, which I describe below, to construct a share of industry materials costs. I also use the 1995 all-establishment shipment series, which provides the domestic-production component of absorption used for China exposure.

BACI CEPII's BACI HS92 version 202401 reports bilateral trade by exporter, importer, product, and year (CEPII, 2024). I use Japan's imports and exports in 1995 and 2018 to construct supplier import penetration. Imports from China also enter China exposure following Autor et al. (2013). BACI also supplies the foreign trade flows used to construct the offshoring instrument. BACI reports trade in current thousand U.S. dollars. I convert these values to current million yen using the annual yen-per-dollar AEXJPUS exchange rate (Board of Governors of the Federal Reserve System, 2026).

Job Tag Job Tag version 4.00.01 reports occupation-level mean responses on a one-to-five scale (Japan Institute for Labour Policy and Training (JILPT), 2023). I construct the routine score from three items: the importance of repeating the same work (IPD_04_05_011), the importance of working at a pace set by equipment (IPD_04_05_012), and the importance of controlling machines and machine-production processes (IPD_04_10_018). I construct the offshorability score from the frequency of face-to-face discussion (IPD_04_05_002) and the degree of physical proximity to other people (IPD_04_05_033). I standardize each item across occupations. The routine score is the equally weighted sum of the three standardized routine items, standardized again to have mean zero and unit standard deviation. The offshorability score is the negative of the standardized sum of the two interaction measures, so a higher value indicates less need for face-to-face discussion and physical proximity. The routine-score construction follows Komatsu and Mugiya (2021). I map Job Tag occupations to the 2005 Japanese occupational classification, average the scores when several Job Tag occupations map to the same occupation, and classify occupations in the highest third of each score as routine and offshorable, respectively.

2.3 Automation Exposure

Let $W_{gj,1995}$ denote the wage bill of group g in JIP industry j , and let $W_{g,1995}$ denote the group's total wage bill. Let r_{gj} be the routine-occupation share of the group–industry wage bill and r_j the corresponding national industry share. The automation exposure weights, industry shocks, and group exposure are

$$\omega_{gj} = \frac{W_{gj,1995}}{W_{g,1995}}, \quad q_{gj}^A = \omega_{gj} \frac{r_{gj}}{r_j}, \quad a_j = -d \log s_j^L, \quad X_g^A = \sum_j q_{gj}^A a_j. \quad (4)$$

I set $q_{gj}^A = 0$ when $r_j = 0$. If v_j^A indicates a valid endpoint shock, then $S_g^A = \sum_j q_{gj}^A v_j^A$ is the automation incomplete-share control (Borusyak et al., 2025).

2.4 Offshoring Exposure

Let $W_{gi,1995}$ denote the wage bill of group g in buyer JSIC3 industry i , and let o_{gi} and o_i denote the group–industry and national industry offshorable-task wage-bill shares. The offshoring exposure weight is

$$\omega_{gi} = \frac{W_{gi,1995}}{W_{g,1995}}, \quad q_{gi}^O = \omega_{gi} \frac{o_{gi}}{o_i}. \quad (5)$$

I set $q_{gi}^O = 0$ when $o_i = 0$.

Let M_{ht} and X_{ht} denote Japan's total BACI imports and exports in supplier JSIC3 industry h , and let Y_{ht} denote CoM shipments. Supplier import penetration is

$$\mu_{ht} = \frac{M_{ht}}{Y_{ht} + M_{ht} - X_{ht}}. \quad (6)$$

The denominator is Japanese absorption. BACI values are converted from current thousand U.S. dollars to current million yen using the same-year exchange rate.

Let $S = \{006, \dots, 059\}$ be the manufacturing JIP products. The denominator additionally includes products 001, 003, 004, and 005, so $D = \{001, 003, 004, 005\} \cup S$.² Let $\tilde{I}_{it}$ be the resulting imported-input purchases and $\tilde{U}_{it}$ the corresponding buyer projection of all purchases in D . The imported-input cost share is

$$m_{it} = \frac{\tilde{I}_{it}}{\tilde{U}_{it}}. \quad (7)$$

²I apply supplier import penetration to the S -use cells, split JIP suppliers across JSIC3 suppliers with CoM shipment weights, and aggregate JIP buyer columns to JSIC3 buyers with JIP output weights.

Table 1: Summary Statistics

	Mean	SD	Min.	Median	Max.	N
1995 log monthly earnings	8.43	0.36	7.40	8.47	9.59	1504
1995 log aggregate hourly wage	3.23	0.37	2.24	3.24	4.58	1504
1995 log monthly hours	5.21	0.06	4.83	5.22	5.35	1504
1995–2018 change in log monthly earnings	-0.07	0.12	-1.13	-0.09	0.85	1504
1995–2018 change in log aggregate hourly wage	-0.06	0.14	-1.24	-0.08	0.96	1504
1995–2018 change in log monthly hours	-0.01	0.03	-0.23	0.00	0.25	1504
Automation exposure	0.08	0.14	-1.92	0.06	2.09	1504
Offshoring exposure	0.03	0.04	-0.02	0.03	0.53	1504
China exposure	0.02	0.02	0.00	0.02	0.17	1504

Notes: This table shows summary statistics for the outcomes and exposures. The columns give the weighted mean, standard deviation, minimum, median, maximum, and number of observations. The sample contains 1,504 prefecture–sex–age–education groups observed in 1995 and 2018, and all statistics use 1995 total hours as analytic weights. Aggregate hourly wages equal survey-weighted total monthly earnings divided by survey-weighted total monthly hours within each group. Automation, offshoring, and China exposure use 1995 industry wage-bill shares and task intensities.

The buyer-industry shock is the log decline in the domestic share of materials costs, and the group exposure is

$$z_i^O = -d \log(1 - m_i), \quad X_g^O = \sum_i q_{gi}^O z_i^O. \quad (8)$$

Positive values represent increased reliance on imported inputs. The treatment uses 147 buyer industries with positive endpoint materials costs and valid domestic shares. If v_i^O indicates a valid shock, the offshoring incomplete-share control is

$$S_g^O = \sum_i q_{gi}^O v_i^O. \quad (9)$$

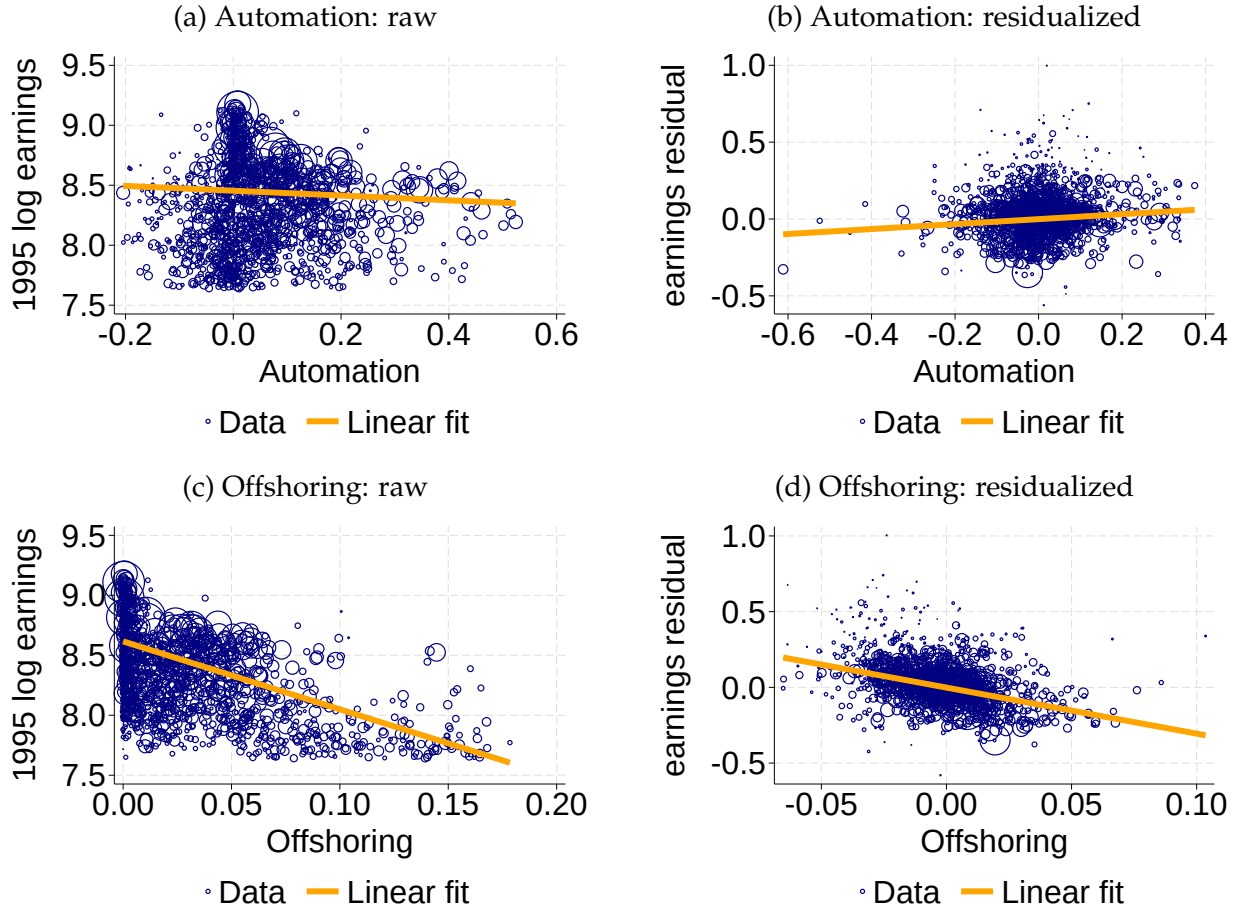
2.5 Summary Statistics

Table 1 summarizes the analysis sample. Between 1995 and 2018, monthly earnings and hourly wages decline on average and at the median, while hours worked per month stay almost constant on average. On average, automation increases by 0.08 log points, and offshoring increases by 0.03 log points. Both measures have large standard deviations of 0.14 and 0.04 log points, respectively, compared to the magnitudes of the means and medians.

2.6 Exposure by Baseline Level

Before I study the changes in earnings by different exposures, I show how exposures vary by the initial levels of earnings. Figure 2 compares each exposure with log monthly earnings in 1995.

Figure 2: Initial monthly earnings, Automation, and Offshoring



Notes: This figure shows log mean real monthly earnings in 1995 against automation in panels (a) and (b) and offshoring in panels (c) and (d). Panels (a) and (c) show raw relationships, and panels (b) and (d) show residualized relationships. For each exposure, the displayed sample drops groups below the hours-weighted 1st percentile or above the 99th percentile of either 1995 log monthly earnings or the exposure. This leaves 1,391 groups in panels (a) and (b) and 1,423 groups in panels (c) and (d); the residualization regressions use all 1,504 groups. Navy circles show the retained groups, circle areas are proportional to 1995 total hours, and orange lines show hours-weighted linear fits. Percentile calculations, residualization, and linear fits use 1995 total hours as analytic weights. Panels (b) and (d) residualize both axes on the exposure-specific incomplete-share control, China exposure and its incomplete-share control, and prefecture, sex, five-year age-group, and college fixed effects.

Here, for visibility, I drop groups below the hours-weighted 1st percentile or above the 99th percentile of either plotted variable. This leaves 1,391 groups in the automation pan-

els and 1,423 groups in the offshoring panels. In the full sample, the conditional automation coefficient is 0.07 (0.06), while the conditional offshoring coefficient is -2.76 (0.43). They imply that the automation exposure is not correlated with the baseline log earnings, while the offshoring exposure is concentrated on groups with ex-ante lower earnings. A one-standard-deviation increase in offshoring is associated with approximately 0.101 log points lower initial earnings.

3 Automation, Offshoring, and Changes in Earnings

3.1 OLS evidence

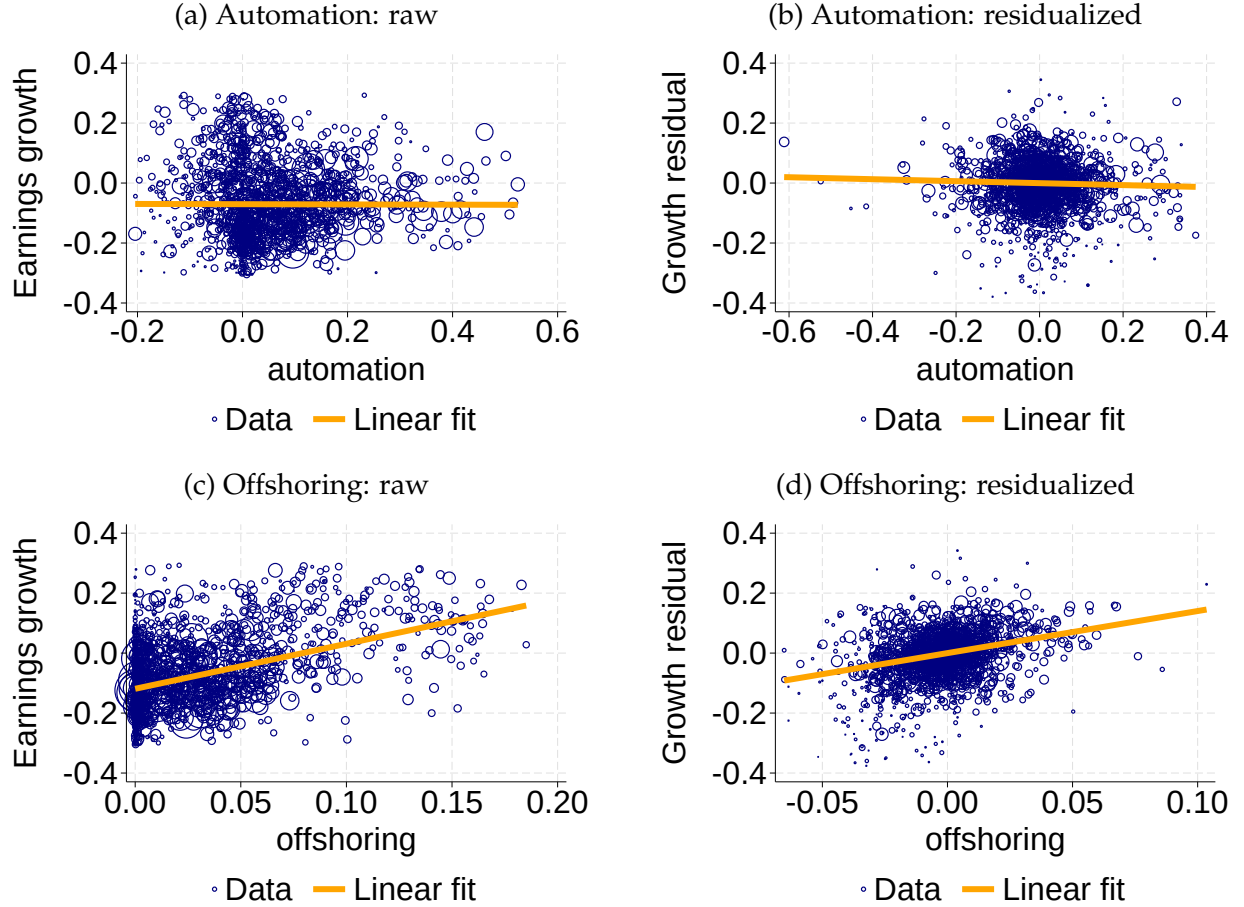
Using the specification in equation (3) and the variables described in Section 2, I begin with the OLS results for automation and offshoring. Figure 3 compares the 1995–2018 change in log monthly earnings with automation and offshoring.

Panels (a) and (b) show that automation has little relationship with earnings growth, in contrast to the U.S. evidence in [Acemoglu and Restrepo \(2022\)](#). Panels (c) and (d) show the results for offshoring. Offshoring has a positive relationship in both the raw and residualized panels.

Table 2 reports OLS estimates corresponding to the visual patterns in Figure 3. In the automation block, the estimate is -0.04 (0.05) with the automation incomplete-share control in column (1). It is -0.02 (0.04) after adding China exposure and its incomplete-share control in column (2), and remains -0.02 (0.03) after adding prefecture and demographic fixed effects in column (3). Multiplying the column (3) coefficient by the 0.14 standard deviation of automation gives an association of only -0.003 log points. These statistically and economically insignificant estimates provide little evidence that groups more exposed to industry labor-share declines experienced systematically different earnings growth.

In the offshoring block, the estimate is 2.45 (0.27) in column (4) and 2.47 (0.37) after adding the China controls in column (5). Adding both sets of fixed effects lowers the estimate to 1.34 (0.27) in column (6). A one-standard-deviation increase in offshoring is therefore associated with approximately 0.049 log points of additional earnings growth.

Figure 3: Changes in monthly earnings, automation, and offshoring



Notes: This figure shows the 1995–2018 change in log mean real monthly earnings against automation in panels (a) and (b) and offshoring in panels (c) and (d). Panels (a) and (c) show raw relationships, and panels (b) and (d) show residualized relationships. For each exposure, the displayed sample drops groups below the hours-weighted 1st percentile or above the 99th percentile of either the earnings change or the exposure. This leaves 1,314 groups in panels (a) and (b) and 1,344 groups in panels (c) and (d); the residualization regressions use all 1,504 groups. Navy circles show the retained groups, circle areas are proportional to 1995 total hours, and orange lines show hours-weighted linear fits. Percentile calculations, residualization, and linear fits use 1995 total hours as analytic weights. Panels (b) and (d) residualize both axes on the exposure-specific incomplete-share control, China exposure and its incomplete-share control, and prefecture, sex, five-year age-group, and college fixed effects.

Table 2: OLS: Changes in log monthly earnings: automation and offshoring

	Automation			Offshoring		
	(1)	(2)	(3)	(4)	(5)	(6)
Exposure	-0.04 (0.05)	-0.02 (0.04)	-0.02 (0.03)	2.45 (0.27)	2.47 (0.37)	1.34 (0.27)
Observations	1504	1504	1504	1504	1504	1504
Incomplete-share control	✓	✓	✓	✓	✓	✓
China exposure control		✓	✓		✓	✓
Prefecture FE			✓			✓
Sex, age, and college FE			✓			✓

Notes: This table shows weighted OLS regressions of the 1995–2018 change in log mean real monthly earnings on automation or offshoring. The first three columns use automation, and the last three use offshoring. Columns (1) and (4) include the treatment-specific incomplete-share control. Columns (2) and (5) add China exposure and its incomplete-share control. Columns (3) and (6) add prefecture, sex, five-year age-group, and college fixed effects. The sample contains 1,504 prefecture–sex–age–education groups. Every specification uses 1995 total hours as analytic weights. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

3.2 2SLS estimates

The results in Figure 3 and Table 2 establish associations between changes in earnings and offshoring, but they do not provide a causal interpretation. An unobserved sector-level demand shock might lead firms to buy more intermediate goods, raise productivity, and pay higher wages. Following the logic of Autor et al. (2013), I construct an instrumental variable for offshoring using foreign trade growth outside Japan and China.³

Instrument Let F_{ht}^{-JC} be sector-total trade from origins other than Japan and China to destinations other than Japan. The foreign growth factor and predicted 2018 Japanese imports are

$$G_h^{-JC} = \begin{cases} F_{h,2018}^{-JC} / F_{h,1995}^{-JC}, & F_{h,1995}^{-JC} > 0, \\ 1, & F_{h,1995}^{-JC} = 0, \end{cases} \quad \hat{M}_{h,2018} = M_{h,1995} G_h^{-JC}. \quad (10)$$

The 1995 Japanese-import base includes imports from every non-Japan origin.

³I do not have a comparable instrument for automation, so the 2SLS analysis focuses on offshoring.

Predicted supplier penetration replaces only the 2018 import component,

$$\hat{\mu}_{h,2018} = \frac{\hat{M}_{h,2018}}{Y_{h,2018} + \hat{M}_{h,2018} - X_{h,2018}}. \quad (11)$$

I pass $\hat{\mu}_{h,2018}$ through the same projected-use formula to obtain $\hat{m}_{i,2018}$. The predicted buyer shock and group instrument are

$$\hat{z}_i^O = \log(1 - m_{i,1995}) - \log(1 - \hat{m}_{i,2018}), \quad Z_g^O = \sum_i q_{gi}^O \hat{z}_i^O. \quad (12)$$

Results Table 3 instruments offshoring with predicted Japanese import growth generated by foreign trade growth outside Japan and China. For monthly earnings, the coefficient is 2.42 (0.58) with the incomplete-share and China controls in column (1). Adding prefecture and demographic fixed effects lowers it to 1.31 (0.39) in column (2). The latter estimate implies 0.048 log points of earnings growth for a one-standard-deviation increase in offshoring. Under the exclusion restriction, these estimates indicate that imported inputs raised earnings for more exposed groups.

3.3 Wages or Hours?

To decompose the changes in monthly earnings, I estimate the same models with two additional outcomes—hourly wages and hours worked. Columns (3) and (4) in Table 3 report the hourly-wage results. They closely mirror the earnings estimates. The coefficient is 2.62 (0.72) in column (3) and 1.40 (0.51) in column (4). The latter corresponds to about 0.051 log points for a one-standard-deviation increase in offshoring.

The hours coefficients are -0.19 (0.14) in column (5) and -0.09 (0.15) in column (6). The latter is approximately -0.003 log points for a one-standard-deviation increase in offshoring and is imprecisely estimated.

In sum, the wage estimates are close to the corresponding earnings estimates, while the hours estimates are statistically and economically insignificant. They suggest that the earnings response did not arise from longer working time.

3.4 Pre-trends

Table 4 uses the 1980–1995 changes in monthly earnings, aggregate hourly wages, and monthly hours while retaining the 1995–2018 offshoring exposure and instrument. Each outcome uses the full-control specification, which includes the incomplete-share and

Table 3: 2SLS: Earnings, hourly wages, and hours: offshoring

	Monthly Earnings		Hourly Wages		Hours Worked	
	(1)	(2)	(3)	(4)	(5)	(6)
Instrumented exposure	2.42 (0.58)	1.31 (0.39)	2.62 (0.72)	1.40 (0.51)	-0.19 (0.14)	-0.09 (0.15)
Observations	1504	1504	1504	1504	1504	1504
Positive-weight industry shocks	144	144	144	144	144	144
1st Stage F-Stat.	89.6	89.6	89.6	89.6	89.6	89.6
Incomplete-share control	✓	✓	✓	✓	✓	✓
China exposure control	✓	✓	✓	✓	✓	✓
Prefecture FE		✓		✓		✓
Sex, age, and college FE		✓		✓		✓

Notes: This table shows weighted 2SLS regressions of the 1995–2018 changes in log mean real monthly earnings, the log aggregate real hourly wage, and log mean monthly hours on instrumented offshoring. The aggregate hourly wage equals survey-weighted total monthly earnings divided by survey-weighted total monthly hours within each group. The instrument predicts 2018 Japanese imports using sector-total foreign trade growth from origins other than Japan and China to destinations other than Japan; its 1995 Japanese-import base includes all non-Japan origins. Columns (1), (3), and (5) include the offshoring incomplete-share control, China exposure, and its incomplete-share control. Columns (2), (4), and (6) add prefecture, sex, five-year age-group, and college fixed effects. The sample contains 1,504 prefecture–sex–age–education groups. Every specification uses 1995 total hours as analytic weights. Exposure-robust standard errors (Borusyak et al., 2025) are in parentheses.

China controls together with prefecture and demographic fixed effects. The earnings, wage, and hours estimates are 0.04 (0.26), 0.12 (0.29), and -0.07 (0.12), respectively. A one-standard-deviation increase in offshoring implies only 0.002 log points of pre-period earnings growth. These estimates provide no evidence of differential pre-trends in any of the three outcomes.

Table 4: 2SLS: Pre-trend check for earnings, hourly wages, and hours; offshoring

	Monthly Earnings	Hourly Wages	Hours Worked
	(1)	(2)	(3)
Instrumented exposure	0.04 (0.26)	0.12 (0.29)	-0.07 (0.12)
Observations	1485	1485	1485
Positive-weight industry shocks	144	144	144
1st Stage F-Stat.	73.3	73.3	73.3
Incomplete-share control	✓	✓	✓
China exposure control	✓	✓	✓
Prefecture FE	✓	✓	✓
Sex, age, and college FE	✓	✓	✓

Notes: This table shows weighted 2SLS regressions of the 1980–1995 changes in log mean real monthly earnings, the log aggregate real hourly wage, and log mean monthly hours on instrumented offshoring. The offshoring exposure and instrument retain their 1995–2018 definitions, including group exposure weights measured in 1995, so this is a retrospective check rather than a fully predetermined pre-trend test. Every column includes the offshoring incomplete-share control, 1995–2018 China exposure and its incomplete-share control, and prefecture, sex, five-year age-group, and college fixed effects. The sample contains 1,485 prefecture–sex–age–education groups observed in both 1980 and 1995. Every specification uses 1980 total hours as analytic weights. Exposure-robust standard errors ([Borusyak et al., 2025](#)) are in parentheses.

4 Conclusion

Earnings differences across groups of full-time regular employees narrowed between 1995 and 2018. The exposure regressions provide little evidence that the theory-guided automation measure contributed to this convergence. Offshoring has a different pattern: groups with greater exposure began with lower earnings and experienced faster earnings growth. The IV estimates indicate that a one-standard-deviation increase in offshoring raised monthly earnings by about 0.048 log points under the stated exclusion restriction.

The response operates almost entirely through aggregate hourly wages rather than monthly hours. This distinction matters because an earnings increase generated by longer hours has a different interpretation from an increase in earnings per hour. The results concern differences between demographic groups and do not measure within-group inequality. They also rely on total labor-share declines as a proxy for automation and on foreign trade growth as an instrument for offshoring. Within those limits, access to imported inputs appears more closely connected to Japan’s between-group earnings convergence than exposure to industry labor-share declines. This pattern contrasts with the U.S. evidence reported by [Acemoglu and Restrepo \(2022\)](#).

Conflict of Interest

I state that there is no conflict of interest.

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